AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T316
Advanced Networking

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21/Apr. /2026

mber of Exam Pages: 3 uding this cover sheet)	Time Allowed: 2 hours

structions:

- Total Marks:**60 marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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PART 1: Multiple Choices Questions (10 x 1 = 10 Marks)

Select the best answer for each question.

1. Which protocol is used for distributing MPLS labels?
   a) RSVP b) LDP c) OSPF d) BGP
2. On which layer of the OSI model does IP operate?
   a) Physical layer b) Data link layer c) Network layer d) Transport layer
3. What is the basic unit of data transmission in IP?
   a) Frame b) Packet c) Segment d) Datagram
4. Which of the following statements correctly describe the differences in frame structures between 10BASE-T, 100BASE-T, and Gigabit Ethernet?
   a) 10BASE-T and 100BASE-T have the same maximum standard frame size of 1518 bytes.
   b) Gigabit Ethernet introduces jumbo frames, which can be up to 9000 bytes in size.
   c) Gigabit Ethernet uses a carrier extension to ensure a minimum frame size for collision detection.
   d) All the above.
5. What is the primary purpose of the ARP (Address Resolution Protocol)?
   a) Assign IP addresses b) Map IP addresses to MAC addresses
   c) Route packets between networks d) Encrypt network traffic
6. Which routing protocol is classified as a link-state protocol?
   a) RIP b) BGP c) OSPF d) EIGRP
7. In the context of network security, what does IPS stand for?
   a) Internet Protocol Security b) Intrusion Prevention System
   b) Internal Packet Scanning d) Integrated Protection Service
8. Which protocol is used for dynamic routing in large enterprise networks and the internet?
   a) RIP b) OSPF c) EIGRP d) BGP
9. Which of the following are the benefits of network virtualization in terms of enhancing network agility and scalability?
   a) Rapid provision of network resources b) Improved hardware utilization
   b) Enhanced network segmentation d) All the above
10. Which of the following best describes the MPLS VPN concept, where customer traffic is separated and transported across a service provider's network?
    a) VLAN tagging b) BGP routing c) MPLS tunneling d) Layer 2 switching

PART 2: Short Answer Questions (20 Marks)

Question 1: (10 marks)

A. Explain how the virtual concatenation scheme works. (4 marks)

B. Why does it have to be implemented only at the originating and terminating nodes? Elaborate three reasons. (6 marks)

Question 2: (10 Marks)

A. Describe the concept of LSP tunnels in MPLS. (3 marks)

B. How does label stacking solve potential forwarding issues at the end of a tunnel? Highlight two. (4 marks)

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C. Identify two main components of the Resource Reservation Protocol (RSVP) and explain their roles in resource allocation. (3 marks)

PART 3: Long/Problem Solving Questions (30 Marks)

Question 1: (10 marks)

A. Calculate the total time required to transfer a 1000-KB file in the case of the bandwidth is 1.5 Mbps, and data packets can be sent continuously, assuming an RTT of 50 ms, a packet size of 1 KB data, and an initial 2×RTT of "handshaking" before data is sent. (5 marks)

B. Calculate the latency (from first bit sent to last bit received) for a 100-Mbps Ethernet with a single store-and-forward switch in the path and a packet size of 12,000 bits. Assume that each link introduces a propagation delay of 10 μs and that the switch begins retransmitting immediately after it has finished receiving the packet. (5 marks)

Question 2: (10 marks)

A. Explain the key operational differences between EGP and IGP. Where is each typically implemented within network architecture? Fill out Table P3Q2. (7 marks)

Aspects	EGP	IGP
Where does it run?	1.5 marks	1.5 marks
Where is it used?	1 mark	1 mark
Examples	1 mark	1 mark

Table P3Q2: Comparison table

B. BGP advertises a route by including the prefix and set of attributes. How does BGP use the AS-PATH and NEXT-HOP attributes? (3 marks)

Question 3: (10 marks)

A. How does DiffServ contribute to achieving QoS on the internet? (4 marks)

B. Compare the DiffServ (Differentiated Services) and IntServ (Integrated Services) QoS models. Discuss their key characteristics in terms of scalability, complexity, traffic handling, and resource allocation. (6 marks)

End of the questions.

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